Reliability of SWOT water surface level measurements for small inland lakes

Space Geodesy Research Project

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Abstract

Monitoring inland water levels is critical for hydrological management and climate-impact assessment, yet global in situ networks remain spatially sparse. Radar altimetry enables consistent water-level observations, and the Surface Water and Ocean Topography (SWOT) mission extends this capability to smaller inland waters. This study presents a modular, lake-independent processing framework to generate lake-wise water surface elevation time series from the SWOT Level 2 Lake Single Pass Vector Product and assesses their reliability over Swiss lakes using daily in situ gauge records. The results show low-centimeter mean offsets and RMSE, along with high correlation between SWOT and in situ measurements, indicating strong agreement.

1 Introduction

Monitoring water levels is essential for managing water resources and assessing climate impacts, but sparse and declining ground-based networks limit global coverage (Fu, Alsdorf, et al. 2009, Palmer, Kutser, and Hunter 2014). Technological advances have shifted water-level monitoring from manual methods to remote sensing, with radar remote sensing enabling frequent, weather-independent observations, including in remote regions. Over the past decades different missions have demonstrated the capability of radar altimetry to measure inland water levels with increasing accuracy; however, spatial coverage remained limited due to nadir-only observation geometry (Biancamaria, Lettenmaier, and T. M. Pavelsky 2025; Fu, T. Pavelsky, et al. 2024; Frappart et al. 2021).

The Surface Water and Ocean Topography (SWOT) mission (*NASA-SWOT* 2025), launched in 2022, represents a major step forward in observing global surface water dynamics. SWOT is equipped with the Ka-band Radar Interferometer (KaRIn), which enables wide-swath measurements of water surface elevation (WSE) and extent through interferometric SAR observations. Inland water bodies are monitored at approximately 100 m resolution with a 21-day repeat cycle(Chen et al. 2018).

The main objective of this study is to develop a modular framework for SWOT-based time series analysis of WSE and surface area using the L2 LakeSP SWOT data product. The framework is applied to a selection of Swiss lakes of various sizes, in order to assess the performance of SWOT on larger and smaller water bodies. The accuracy of the SWOT data is evaluated using a combination of time series and statistical metrics that quantify its agreement with the in situ measurements.

2 Data

As data products, the SWOT Level 2 KaRIn High Rate Lake Single Pass Vector Product (L2_HR_LakeSP), versions C and D was used. Version C provided a more comprehensive base for the SWOT data analysis, while version D was more robust but incomplete for the two-year analysis time frame. The LakeSP product provides lake-aggregated estimates of WSE and

surface area, along with associated uncertainty descriptors and quality flags. Separate datasets were generated for each lake using lake IDs obtained from *HydroWeb* 2025.

Daily mean lake-level measurements were obtained from BAFU (Federal Office for the Environment) for the selected validation stations. These records serve as a reference for evaluating SWOT-derived WSE.

3 Methods

A modular Python framework was developed to generate lake-wise WSE and area time series from SWOT LakeSP products. The lake-independent workflow transforms datasets into comparable time series, removes outliers, and analyses agreement with *in situ* observations.

3.1 Reference system harmonization

To compare the datasets, both in-situ and SWOT heights were first transformed to a common reference frame using PROJ (Open Source Geospatial Foundation 2025). In-situ heights were originally referenced to LN02 (EPSG:5728) based on the CHGeo2004 geoid, while SWOT heights were expressed using the EGM2008 geoid. Since water movement is governed by the Earth's gravity field, geoid undulation differences between CHGeo2004 and EGM2008 were computed at each measurement station, after transforming their coordinates to the target reference frame.

3.2 SWOT data cleaning and quality control

The processing of SWOT observations is designed to ensure that only physically plausible, low-uncertainty measurements are included in the analysis. After the analysis, a unified daily SWOT time series was constructed by prioritizing Version D when both versions were available on the same day.

Attribute based filtering: To ensure high-quality data, raw measurements were filtered using the product's quality attributes. Measurements flagged as bad or suspicious were removed using the `quality_f` flag, and surface values not covering the entire lake were excluded using the `partial_f` flag.

Value-domain filtering: WSE measurements were screened using thresholds derived from a preliminary analysis of Swiss lake dynamics. This analysis established general ranges and rates of change, rather than lake-specific filters. A WSE range filter removed values deviating more than twice the mean maximum deviation from the median WSE. Similarly, consecutive-difference filtering used interval-based thresholds set to twice the 90th percentile of short-term WSE changes, averaged across lakes.

Uncertainty-based filtering: Uncertainty filtering used the within-lake standard deviation, setting the threshold at the 90th percentile of each lake's values and averaging across lakes to avoid bias from dense sampling or atypical variability. This approach removes only the observations with the highest uncertainty, while retaining most of the dataset.

3.3 Validation metrics

Two complementary analyses were conducted to evaluate the accuracy of SWOT-derived WSE against in situ series for each lake and product version. First, individual observation accuracy was assessed using a pointwise comparison of same-day measurements, considering mean offset, standard deviation, RMSE, and Pearson correlation.

Second, to capture broader temporal patterns despite uneven sampling of SWOT, days with both measurements were linearly interpolated and smoothed using a 30-day low-pass filter. The same metrics were then computed on the smoothed series to assess agreement in long-term dynamics.

4 Results

4.1 Combined time series

The pipeline developed for this project, produces time series of the SWOT WSE measurements that can be layered on top of the verifications measurements as is shown in figure 1. In the same figure, the results of the interpolation are shown and compared to the in situ data.

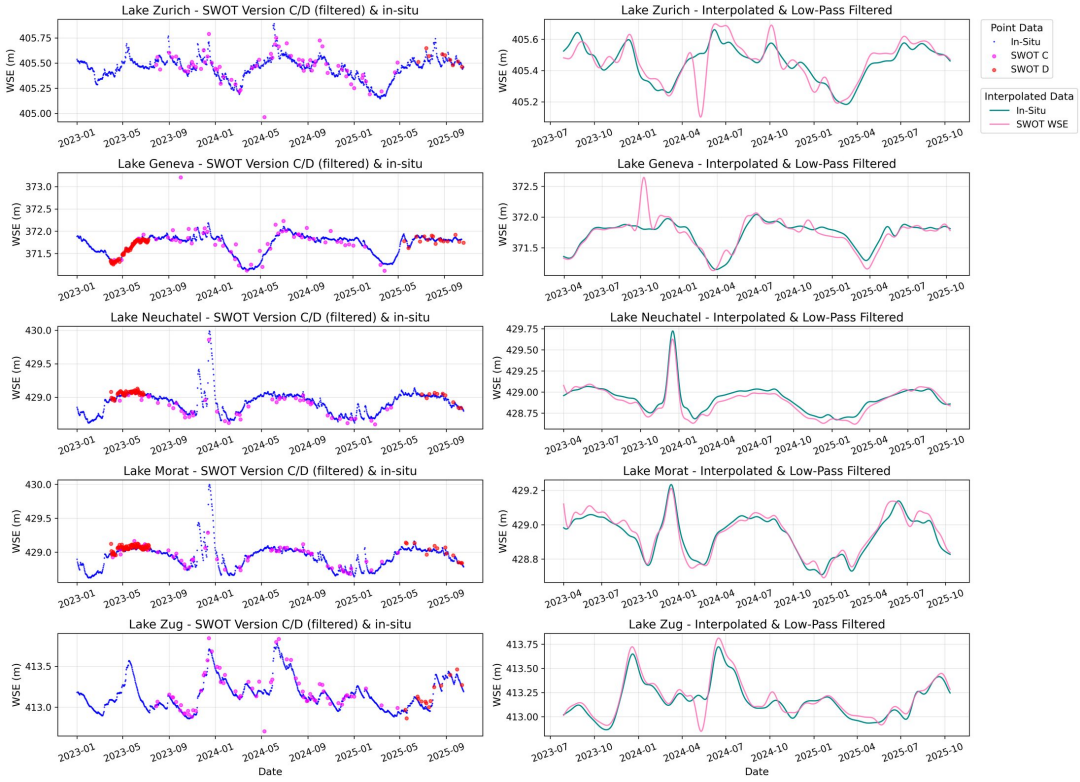


Figure 1: On the left: combined time series of the in situ measurements and SWOT measurements. On the right: plots of the interpolated SWOT time-series for each lake.

In figure 1, three lakes show an agglomeration of SWOT measurements during the spring of 2023. During this period of time the SWOT mission was conducting a fast sampling phase with a one day repeat phase for calibration (Chen et al. 2018). The orbit for this fast repeat cycle sampled only the western part of Switzerland, which is why this unusual density of measurements is not seen in the two eastern lakes of Zug and Zürich.

4.2 Reliability of SWOT WSE measurements

Across the analysed Swiss lakes, SWOT LakeSP data show generally strong agreement with BAFU *in situ* WSE (See table 1).

4.3 SWOT-derived water surface area estimates

A first analysis of the SWOT LakeSP area time series revealed that certain overpasses over bigger lakes (Zürich, Geneva or Neuchâtel) did not observe the whole lake area and were not suitable for area measurements. After the removal of the compromised measurements, plausible time series were obtained for the water surface area and are included in the Appendix 7. The correlation between the WSE and surface area was then computed and is shown in table 2. Note that lake Geneva has no points where the whole lake area was covered. This also correlates with *HydroWeb* 2025, where also no lake area value is available for lake Geneva.

Table 1: Statistics for the reliability of SWOT WSE measurements from the SP Lake Vector product versions C and D with respect to in situ measurements.

	Version C			Version D		
	mean offset [cm]	σ [cm]	correlation	mean offset [cm]	σ [cm]	correlation
Lake Zürich	0.63	9.42	0.7586	2.14	2.54	0.9034
Lake Geneva	3.48	21.91	0.7547	−2.47	2.79	0.9896
Lake Zug	2.98	8.26	0.0872	2.74	3.92	0.9721
Lake Neuchâtel	−5.16	3.19	0.9873	2.12	2.69	0.8784
Lake Morat	0.91	4.01	0.9588	3.19	2.90	0.8819

Table 2: Correlation between SWOT WSE and lake surface area.

Metric	**Lake Zurich**	**Lake Geneva**	**Lake Morat**	**Lake Neuchâtel**	**Lake Zug**
Correlation (C&D)	0.1682	-	0.2991	0.1983	0.1232
p-value (C&D)	0.3196	-	0.0004	0.4157	0.2794

5 Discussion

The results shown in section 4 support that the pipeline developed for this project can reliably produce WSE time series from the SWOT L2-LakeSP product. The produced time series show good point by point correlation with the in situ measurements. The mean offset and the standard deviation (figure 1) remain in the cm range and are in good agreement with what was found in Bazzi et al. 2025. The integrated time series (figure 1) illustrate the possibility to reconstruct the dynamics of the water surface elevation with good accuracy. This result further supports the reliability of the SWOT measurements for inland lakes. Additional complexity is revealed in the analysis of the water surface area, but no obvious connection to the WSE was made visible through a simple correlation analysis.

Overall, these findings indicate that in regions with good in situ coverage, such as Switzerland, the additional value gained from SWOT WSE measurements is negligible. However, these regions can serve as testing ground for pipelines such as the one presented in this paper, which can then be applied to obtain key insight in regions were the in situ coverage is limited. For surface area estimations, the SWOT data product could still contribute significantly even in regions with a dense measuring network, since these measurements are much harder to make from the ground, given that the SWOT measurement fully covered the lake and not partially.

6 Conclusion and Outlook

A modular framework was presented for generating lake WSE and area time series from SWOT L2_HR_LakeSP data and validating the results against daily BAFU gauge measurements over Swiss lakes. Same-day comparisons show strong agreement between SWOT and in situ WSE, demonstrating that SWOT LakeSP provides reliable WSE time series when robust filtering and datum harmonization are applied.

There are many directions that can be explored by future research building on this pipeline. The water surface area product can be tested for it's own reliability or be used to further filter the WSE data to obtain better accuracy. Once the SWOT mission provides longer observation periods, extracting seasonal trends in the time series should also be investigated in order to provide insights into the evolution of hydrospheres around the world.

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Declaration of the Use of AI-based Tools

Table 3 lists all AI tools used during this assignment. After using these services, the authors reviewed and revised the content as necessary and take full responsibility for the submitted code and report.

AI-based Tool	Use Case
ChatGPT	- Spell and grammer check for proper formulations with subsequent editing of the responses - Assisting for LaTeX for figures, sub-figures and tables
GitHub Copilot	- Assitance for the implementations
DeepL	- Translating text from German to English

Table 3: Declaration of the AI-based tools used during this assignment.

7 Appendix

7.1 WSE Time series

In situ time series

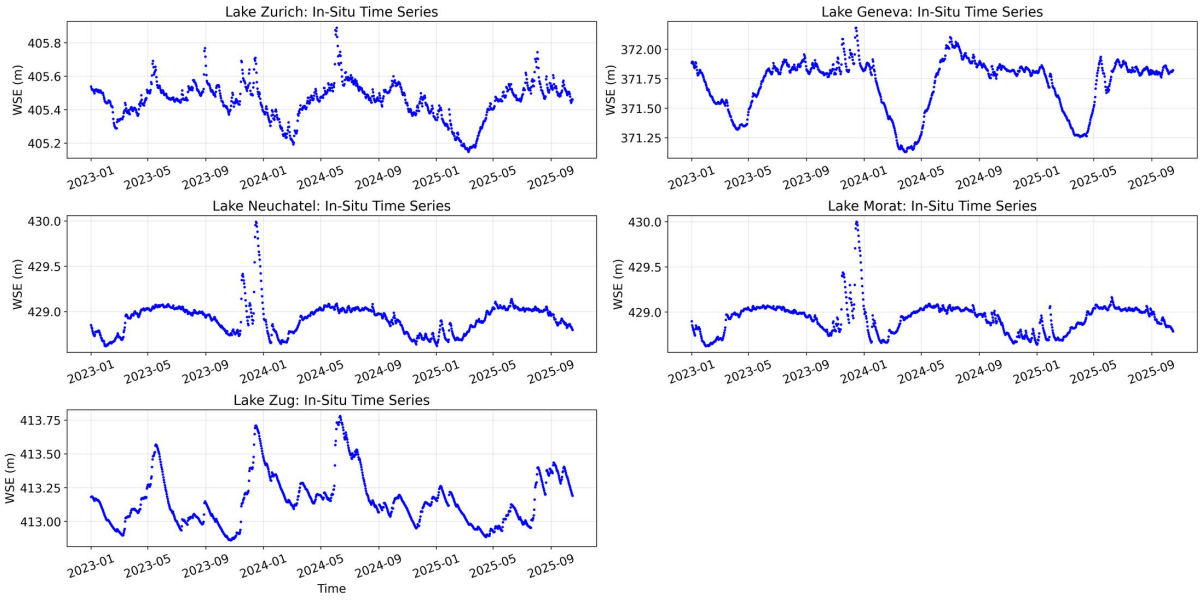


Figure 2: Time series of the in situ wse measurements for the 5 investigated lakes

SWOT unfiltered time series



Figure 3: Time series of the SWOT wse measurements before any filtering was applied for the 5 investigated lakes. Strong outliers (more than (10m) can be seen in all big lakes and smaller but still clear outliers are found in the two smaller lakes

SWOT filtered time series



Figure 4: Time series of the filtered SWOT wse measurements for the 5 investigated lakes

7.2 Water surface area measurements

Raw surface area data

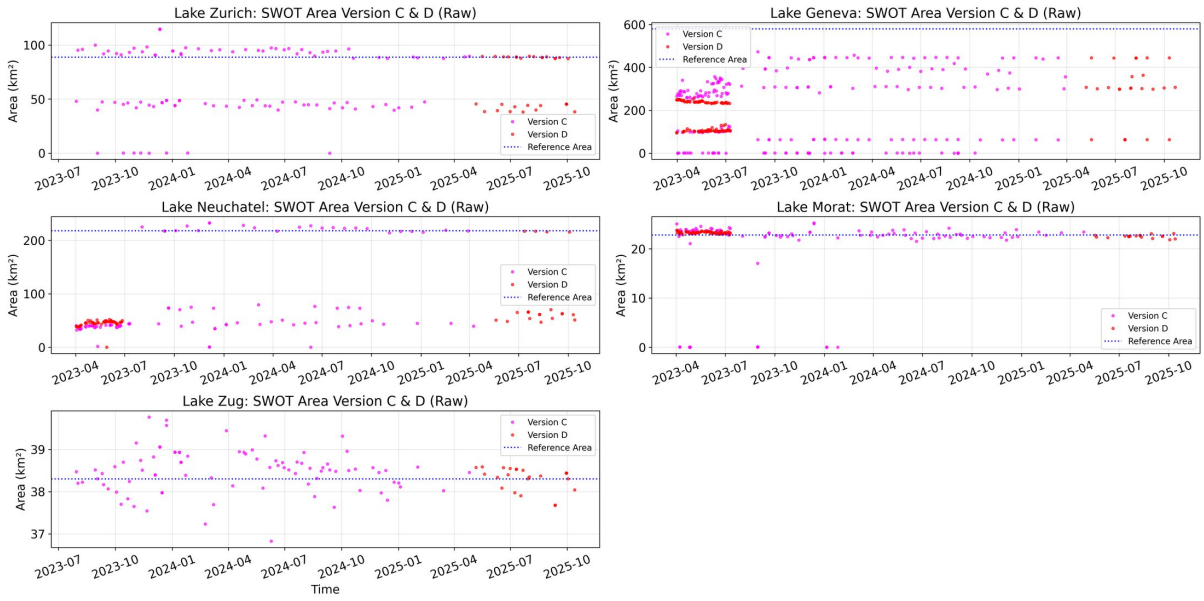


Figure 5: Raw values for the Water surface area for the 5 investigated lakes. Version C and D are shown in different colors and the blue dotted line shows a reference value of the lake surface area. The different lies around whih the points are lustered can be recognized clearly in the bigger lakes such as Zürich, Geneva and Neuchâtel

Filtered data

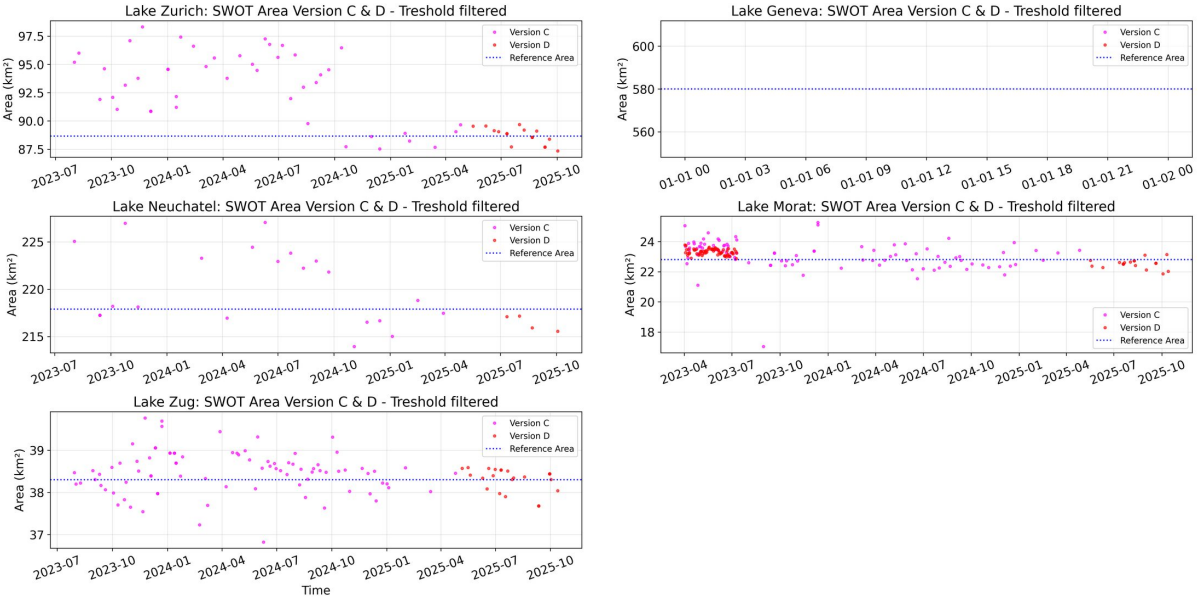


Figure 6: Values for the water surface area after removal of the "half-lake" measurements. The results are now much closer to a single average and could be used for further investigation.